

**HANKO TVARMINNE, Finland**WMO: **027500**Lat: **59.8438N**Lon: **23.2485E**Elev: **3**StdP: **101.29**Time Zone: **2.00 (EUE)**Period: **09-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-----|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |     |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |     |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p) |
| 1             | -17.4      | -14.6 | -19.6                         | 0.7 | -17.1 | -16.9 | 0.8 | -13.6 | N/A                   | N/A  | N/A | N/A  | N/A                   | N/A  | N/A |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 6.2                    | 24.5            | 20.1 | 23.0 | 19.0 | 21.6 | 17.9 | 20.9                | 23.8 | 19.8 | 22.4 | 18.6 | 20.9 | N/A                  | N/A  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 19.7                            | 14.4 | 22.8 | 18.7 | 13.5 | 21.6 | 17.5 | 12.6 | 20.3 | 60.3          | 23.8 | 56.6 | 22.4 | 52.6 | 21.0 | 23.1              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| N/A               | N/A  | N/A | -18.3                      | 26.4 | 4.4                | 2.4 | -21.5                                              | 28.1 | -24.1      | 29.5 | -26.6      | 30.8 | -29.8      | 32.6 |  |  |
|                   |      |     | -18.6                      | 21.5 | 4.4                | 1.5 | -21.7                                              | 22.7 | -24.3      | 23.6 | -26.8      | 24.4 | -30.0      | 25.5 |  |  |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

**Monthly Climatic Design Conditions**

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 6.6    | -3.6  | -3.3  | -0.6  | 3.8   | 9.8   | 14.1  | 17.8  | 16.9  | 13.0  | 7.0   | 3.7   | 0.2  |
|                                                                                  | DBStd       | 8.19   | 4.96  | 5.45  | 3.79  | 2.75  | 3.75  | 2.55  | 2.77  | 2.25  | 2.60  | 3.17  | 3.65  | 4.40 |
|                                                                                  | HDD10.0     | 1962   | 422   | 371   | 330   | 188   | 51    | 1     | 0     | 0     | 7     | 100   | 188   | 305  |
|                                                                                  | HDD18.3     | 4318   | 680   | 604   | 588   | 437   | 264   | 130   | 46    | 57    | 159   | 351   | 438   | 563  |
|                                                                                  | CDD10.0     | 727    | 0     | 0     | 0     | 1     | 44    | 123   | 241   | 213   | 98    | 7     | 0     | 0    |
|                                                                                  | CDD18.3     | 42     | 0     | 0     | 0     | 0     | 1     | 2     | 28    | 12    | 0     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 107    | 0     | 0     | 0     | 0     | 0     | 4     | 86    | 17    | 0     | 0     | 0     | 0    |
|                                                                                  | CDH26.7     | 4      | 0     | 0     | 0     | 0     | 0     | 0     | 4     | 1     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Precipitation                                                                    | PrecAvg     | 677    | 60    | 43    | 38    | 35    | 32    | 51    | 60    | 84    | 55    | 74    | 78    | 66   |
|                                                                                  | PrecMax     | 912    | 107   | 105   | 92    | 60    | 71    | 108   | 173   | 182   | 132   | 186   | 180   | 156  |
|                                                                                  | PrecMin     | 496    | 8     | 3     | 8     | 5     | 5     | 12    | 4     | 8     | 18    | 21    | 7     | 20   |
|                                                                                  | PrecStd     | 97     | 25    | 26    | 18    | 15    | 17    | 24    | 38    | 42    | 29    | 38    | 36    | 32   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 4.7   | 5.3   | 8.3   | 15.0  | 22.1  | 23.2  | 27.1  | 25.2  | 19.5  | 14.2  | 10.4  | 8.4  |
|                                                                                  |             | MCWB   | 4.0   | 2.7   | 4.9   | 8.6   | 14.8  | 16.7  | 20.1  | 20.9  | 16.6  | 12.3  | 9.6   | 7.3  |
|                                                                                  | 2%          | DB     | 4.0   | 3.4   | 6.2   | 12.0  | 19.2  | 21.2  | 25.6  | 23.4  | 18.3  | 13.1  | 9.5   | 7.3  |
|                                                                                  |             | MCWB   | 3.3   | 2.4   | 3.3   | 7.7   | 13.4  | 16.1  | 20.5  | 19.6  | 15.9  | 11.9  | 9.0   | 6.4  |
|                                                                                  | 5%          | DB     | 3.2   | 2.7   | 4.7   | 9.7   | 17.5  | 19.7  | 24.1  | 21.9  | 17.5  | 12.3  | 8.7   | 6.4  |
|                                                                                  |             | MCWB   | 2.7   | 2.0   | 2.5   | 5.9   | 13.0  | 15.5  | 20.0  | 18.5  | 15.5  | 11.2  | 8.0   | 5.5  |
|                                                                                  | 10%         | DB     | 2.3   | 2.1   | 3.5   | 7.9   | 15.9  | 18.6  | 22.7  | 20.7  | 16.7  | 11.5  | 8.0   | 5.4  |
|                                                                                  |             | MCWB   | 1.8   | 1.6   | 2.0   | 5.1   | 12.2  | 14.7  | 19.0  | 17.7  | 14.9  | 10.4  | 7.3   | 4.5  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 4.3   | 3.3   | 5.3   | 9.1   | 15.8  | 18.7  | 22.1  | 21.7  | 17.4  | 12.9  | 10.0  | 7.7  |
|                                                                                  |             | MCDB   | 4.5   | 4.3   | 8.0   | 13.5  | 19.5  | 20.8  | 24.8  | 24.5  | 18.7  | 13.4  | 10.3  | 8.2  |
|                                                                                  | 2%          | WB     | 3.5   | 2.6   | 3.8   | 7.9   | 14.6  | 17.2  | 21.3  | 20.2  | 16.6  | 12.3  | 9.1   | 6.6  |
|                                                                                  |             | MCDB   | 3.9   | 3.2   | 5.4   | 11.8  | 17.9  | 20.0  | 24.3  | 22.6  | 17.9  | 12.8  | 9.4   | 7.1  |
|                                                                                  | 5%          | WB     | 2.8   | 2.2   | 3.1   | 6.6   | 13.6  | 16.2  | 20.6  | 19.2  | 15.9  | 11.6  | 8.2   | 5.6  |
|                                                                                  |             | MCDB   | 3.2   | 2.6   | 4.2   | 9.2   | 16.7  | 18.9  | 23.6  | 21.4  | 17.1  | 12.1  | 8.6   | 6.2  |
|                                                                                  | 10%         | WB     | 1.9   | 1.6   | 2.3   | 5.5   | 12.4  | 15.3  | 19.4  | 18.2  | 15.2  | 10.6  | 7.4   | 4.6  |
|                                                                                  |             | MCDB   | 2.3   | 2.1   | 3.2   | 7.4   | 15.5  | 17.9  | 22.2  | 20.1  | 16.4  | 11.3  | 7.9   | 5.3  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 4.5   | 4.3   | 5.2   | 5.7   | 6.4   | 6.2   | 6.2   | 5.6   | 4.8   | 4.2   | 3.1   | 3.8  |
|                                                                                  |             | MCDBR  | 3.2   | 3.6   | 5.8   | 8.5   | 8.4   | 8.0   | 7.4   | 6.7   | 5.6   | 4.4   | 3.3   | 3.9  |
|                                                                                  |             | MCWBR  | 3.1   | 3.1   | 3.8   | 5.3   | 4.6   | 4.4   | 3.9   | 3.8   | 3.8   | 3.8   | 3.3   | 3.9  |
|                                                                                  | 5% WB       | MCDBR  | 3.3   | 3.4   | 4.9   | 7.8   | 7.4   | 7.0   | 6.5   | 5.6   | 4.8   | 3.9   | 3.2   | 3.8  |
|                                                                                  |             | MCWBR  | 3.3   | 3.0   | 3.6   | 5.0   | 4.6   | 4.1   | 3.8   | 3.8   | 3.6   | 3.8   | 3.3   | 3.9  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.280  | 0.313 | 0.334 | 0.373 | 0.367 | 0.363 | 0.390 | 0.383 | 0.352 | 0.335 | 0.311 | 0.266 |      |
|                                                                                  | taud        | 2.158  | 2.221 | 2.288 | 2.317 | 2.415 | 2.467 | 2.415 | 2.424 | 2.495 | 2.404 | 2.219 | 2.093 |      |
|                                                                                  | Ebn at Noon | 488    | 679   | 793   | 826   | 861   | 870   | 833   | 809   | 778   | 659   | 445   | 349   |      |
|                                                                                  | Edh at Noon | 44     | 71    | 91    | 106   | 103   | 100   | 103   | 95    | 75    | 60    | 42    | 31    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.24   | 0.77  | 2.29  | 3.96  | 5.53  | 5.84  | 5.51  | 4.34  | 2.64  | 1.15  | 0.31  | 0.15  |      |
|                                                                                  | RadStd      | 0.04   | 0.14  | 0.27  | 0.41  | 0.54  | 0.38  | 0.47  | 0.43  | 0.32  | 0.16  | 0.06  | 0.02  |      |

**Historical Trends**

|                                     | DBAvg        | Heating    |            |            | Cooling    |            |            | Degree-Days |            |            |            |
|-------------------------------------|--------------|------------|------------|------------|------------|------------|------------|-------------|------------|------------|------------|
|                                     |              | 99% DB     |            | 99% DP     | 1% DB      | 1% WB      | 1% DP      | HDD10.0     | HDD18.3    | CDD10.0    | CDD18.3    |
|                                     |              | (d)        | (e)        | (f)        | (g)        | (h)        | (i)        | (j)         | (k)        | (l)        | (m)        |
| (46) <b>Station Only</b>            | <b>N/A</b>   | <b>N/A</b> | <b>N/A</b> | <b>N/A</b> | <b>N/A</b> | <b>N/A</b> | <b>N/A</b> | <b>N/A</b>  | <b>N/A</b> | <b>N/A</b> | <b>N/A</b> |
| (47) <b>Regional (60 neighbors)</b> | <b>+0.40</b> | <b>N/A</b> | <b>N/A</b> | <b>N/A</b> | <b>N/A</b> | <b>N/A</b> | <b>N/A</b> | <b>-134</b> | <b>N/A</b> | <b>N/A</b> | <b>N/A</b> |

Nomenclature: See separate page